

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

print("All libraries imported successfully!")
```

All libraries imported successfully!

```
import pandas as pd
url = "https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv"

df = pd.read_csv(url)
print("Dataset loaded successfully!")
```

Dataset loaded successfully!

```
df.head()
```

```
-----
NameError                                Traceback (most recent call last)
/tmp/ipykernel_1876/964094849.py in <cell line: 0>()
----> 1 df.head()

NameError: name 'df' is not defined
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype
---  -
 0   PassengerId    891 non-null    int64
 1   Survived       891 non-null    int64
 2   Pclass        891 non-null    int64
 3   Name          891 non-null    object
 4   Sex           891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
10   Cabin        204 non-null    object
11   Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

```
df.describe()
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

```
print("Number of rows:", df.shape[0])
print("Number of columns:", df.shape[1])
```

Number of rows: 891
Number of columns: 12

```
print(df.isnull().sum())

df["Age"] = df["Age"].fillna(df["Age"].median())
print("Missing Age values after filling:",
      df["Age"].isnull().sum())
```

```
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age           177
SibSp          0
Parch          0
Ticket         0
Fare           0
Cabin         687
Embarked       2
dtype: int64
Missing Age values after filling: 0
```

```
df.drop("Cabin", axis=1, inplace=True)

print("Cabin column removed successfully!")
```

Cabin column removed successfully!

```
print("Number of duplicate records:",
      df.duplicated().sum())
```

Number of duplicate records: 0

```
df.drop_duplicates(inplace=True)

print("Duplicates removed successfully!")
```

Duplicates removed successfully!

```
print(df["Sex"].unique())
print(df["Embarked"].unique())
```

```
['male' 'female']
['S' 'C' 'Q' nan]
```

```
#label encoding
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
df["Sex"] = le.fit_transform(df["Sex"])
print(df["Sex"].head())

#one hot encoding
df = pd.get_dummies(df, columns=["Embarked"], drop_first=True)
print(df.head())
```

```
0    1
1    0
2    0
3    0
4    1
Name: Sex, dtype: int64
   PassengerId  Survived  Pclass  \
0             1         0       3
1             2         1       1
2             3         1       3
3             4         1       1
4             5         0       3

   Name      Sex  Age  SibSp  Parch  \
0  Braund, Mr. Owen Harris    1  22.0      1     0
1  Cumings, Mrs. John Bradley (Florence Briggs Th...    0  38.0      1     0
2    Heikkinen, Miss. Laina    0  26.0      0     0
3  Futrelle, Mrs. Jacques Heath (Lily May Peel)    0  35.0      1     0
4    Allen, Mr. William Henry    1  35.0      0     0

   Ticket    Fare  Embarked_Q  Embarked_S
0  A/5 21171  7.2500      False      True
1    PC 17599 71.2833      False      False
2  STON/O2. 3101282  7.9250      False      True
```

3	113803	53.1000	False	True
4	373450	8.0500	False	True

```
#box plot
import matplotlib.pyplot as plt
import seaborn as sns
plt.figure(figsize=(8,5))
sns.boxplot(x=df["Age"])

plt.title("Box Plot of Age")
plt.show()

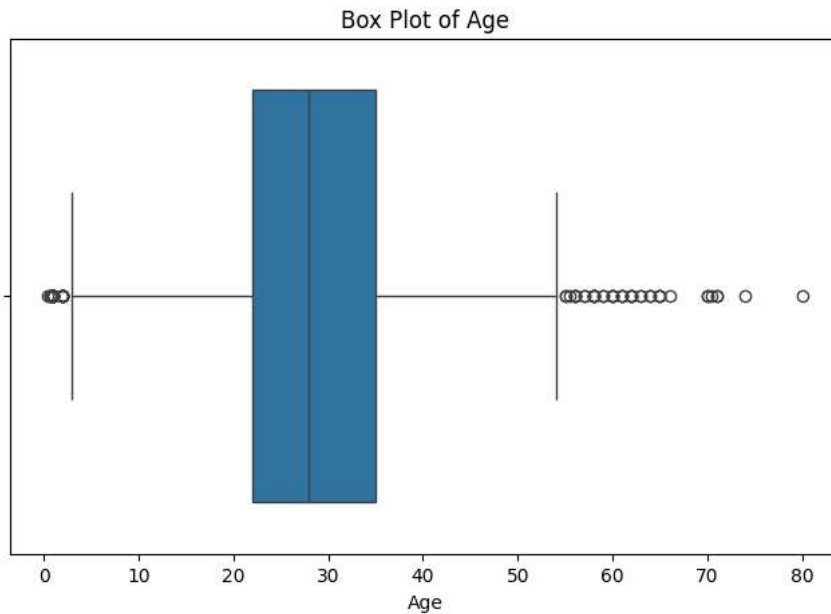
#IQR Calculate
Q1 = df["Age"].quantile(0.25)
Q3 = df["Age"].quantile(0.75)

IQR = Q3 - Q1

print("Q1 =", Q1)
print("Q3 =", Q3)
print("IQR =", IQR)

#lower and upper bound
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

print("Lower Bound =", lower_bound)
print("Upper Bound =", upper_bound)
```



```
Q1 = 22.0
Q3 = 35.0
IQR = 13.0
```

```
outliers = df[
    (df["Age"] < lower_bound) |
    (df["Age"] > upper_bound)
]

print("Number of outliers:", len(outliers))
```

```
Number of outliers: 66
```

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

df[["Age", "Fare"]] = scaler.fit_transform(
    df[["Age", "Fare"]]
)

print(df[["Age", "Fare"]].head())
```

	Age	Fare
0	-0.565736	-0.502445
1	0.663861	0.786845
2	-0.258337	-0.488854
3	0.433312	0.420730
4	0.433312	-0.486337

```
#feature engineering
df["FamilySize"] = df["SibSp"] + df["Parch"] + 1
print(df[["SibSp", "Parch", "FamilySize"]].head())

#Age Group
df["AgeGroup"] = pd.cut(
    df["Age"],
    bins=[0, 12, 18, 60, 100],
    labels=["Child", "Teenager", "Adult", "Senior"]
)
df["FamilySize"] = df["SibSp"] + df["Parch"] + 1
print(df[["Age", "AgeGroup", "FamilySize"]].head())
```

	SibSp	Parch	FamilySize
0	1	0	2
1	1	0	2
2	0	0	1
3	1	0	2
4	0	0	1

	Age	AgeGroup	FamilySize
0	-0.565736	NaN	2
1	0.663861	Child	2
2	-0.258337	NaN	1
3	0.433312	Child	2
4	0.433312	Child	1

```
df.to_csv(
    "titanic_cleaned_preprocessed.csv",
    index=False
)

print("Cleaned dataset saved successfully!")
```

Cleaned dataset saved successfully!